Institutional Ownership Networks and Return Co-movement During Market Stress

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Research Question

Do mega-cap stocks with similar institutional ownership structures exhibit stronger return co-movement during periods of market stress?

Underlying Mechanism:

During volatility shocks, institutional investors may reduce equity exposure or rebalance portfolios toward perceived safer assets

If the same institutions hold multiple firms, these adjustments can generate common trading flows, causing those firms' prices to move together.

Hypothesis:

Firms that share more institutional investors or are more central in the institutional ownership network will exhibit higher return correlations during market stress.

Description of the Topic

- My project examines whether mega-cap firms with similar institutional investor ownership structures move together during periods of market stress.
- Using institutional holdings data from 13F filings, I constructed an ownership network linking firms that share institutional investors.
- I then tested whether firms with greater institutional ownership overlap exhibit stronger return correlations during high-volatility market periods.

Key Terms

- Mega-cap firms: Public companies with extremely large market capitalization (total market value of their equity)
  a. The 18 largest firms were selected to focus on systemically important companies that are widely held by institutional investors while keeping the network interpretable
- Institutional investors: Professional asset managers (mutual funds, pension funds, asset managers) that invest large pools of capital on behalf of clients
  a. Examples include Vanguard, Fidelity, and BlackRock.
- 13F filing: A quarterly SEC disclosure where institutional investors with over $100 million in assets report their U.S. equity holdings
- Daily volatility: The magnitude of a firm's daily stock return movements
- Market volatility: the magnitude of fluctuations in the overall stock market return (measured using the CRSP value-weighted market index)

Data

Data Source	Sample Size	Time Period	Collection Method	Key Variables Created
Yahoo Finance Screener	List of largest U.S. mega-cap firms	Current market snapshot	Web scraping (BeautifulSoup)	Firm tickers
WRDS – CRSP Daily Stock Data	Daily observations for selected mega-cap firms	Study period used in analysis	WRDS SQL query	Daily firm returns, daily market returns, firm identifiers (PERMNO/CUSIP), quarter variable created from date
WRDS – 13F Institutional Holdings (Thomson/Refinitiv S34)	Institutional holdings across institutions and firms	Same study period	WRDS SQL query	Institutional ownership %, shares held, institutional identifier (mgrno), ownership by quarter
FRED API – S&P 500 Data	Daily S&P 500 observations	Same study period	FRED API	Stress days

Data Cleaning Takeaway

Challenge:

Institutional ownership (13F) is reported quarterly, while stock returns and volatility measures are daily.

I initially created quarter start and end date variables to align the datasets.

This added unnecessary complexity and led to mixing up quarter boundaries with volatility dates, producing merged tables with missing values.

Solution:

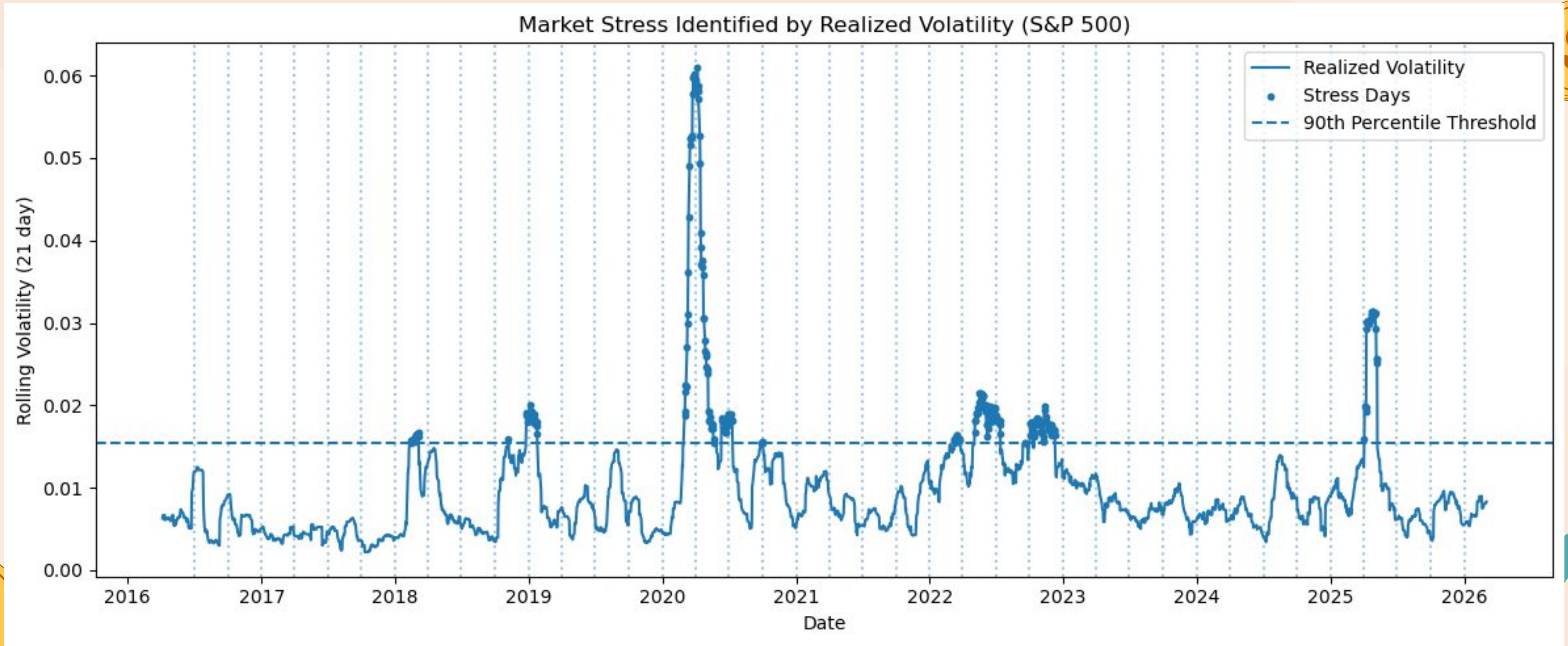
I simplified the merge by using the 13F filing/reporting date instead of storing multiple quarter boundary variables.

Lesson:

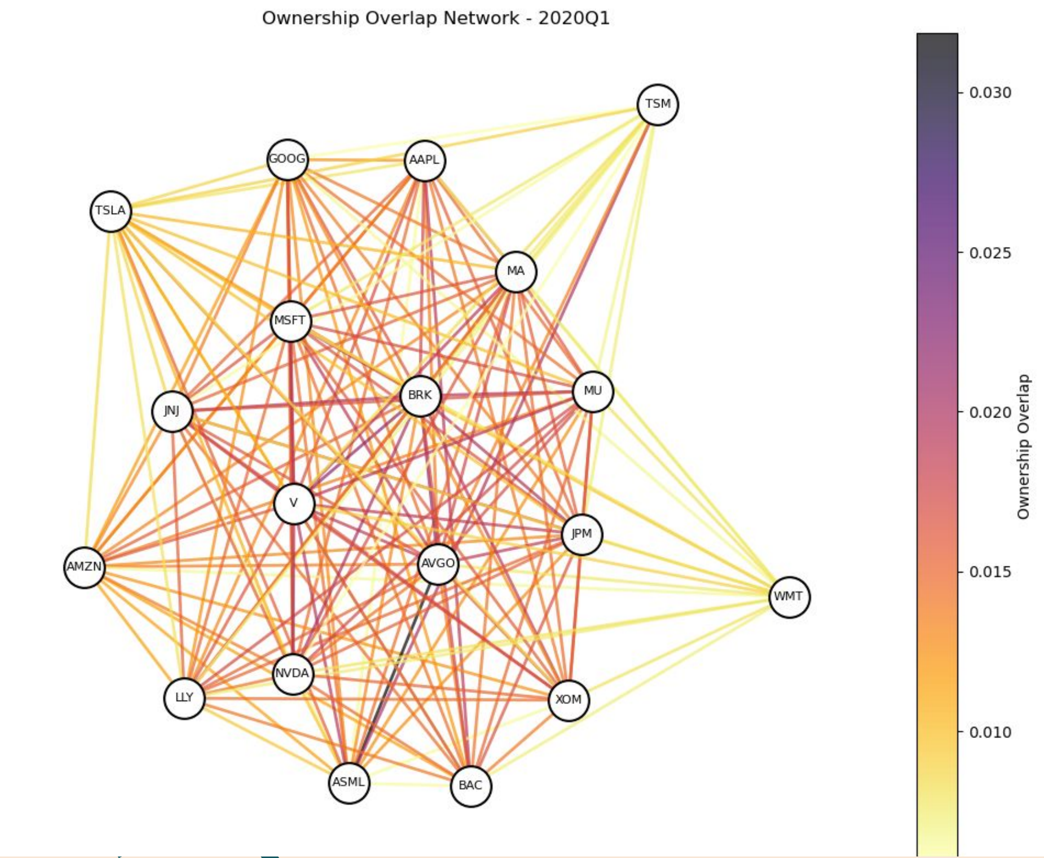
When merging datasets with different time frequencies, it is important to keep track of exactly which time variables are used for each dataset and avoid adding unnecessary variables that complicate merges.

Stress Days

Stress days = days falling within the top 10% of market volatility from 2016 to 2026



Ownership Overlap Network– 2020Q1



What it shows

- Network of firms connected by shared institutional investors for one quarter
- Nodes: firms (tickers)
- Edges: overlap in institutional ownership
- Edge color: strength of ownership overlap

How it was built

- Created a firm × institution ownership matrix
- Calculated firm overlap using $W \times W^T$ where W = ownership percentages
- Filtered weak connections (overlap < 0.006) to reduce clutter
- Built and visualized the network using NetworkX + Matplotlib
- Node positions generated with a spring layout

Descriptive Measures

Ownership Overlap Summary

mean	0.012826
std	0.004415
min	0.001916
25%	0.008999
50%	0.013841
75%	0.016038
max	0.036166

Network Structure Statistics

Nodes: 19

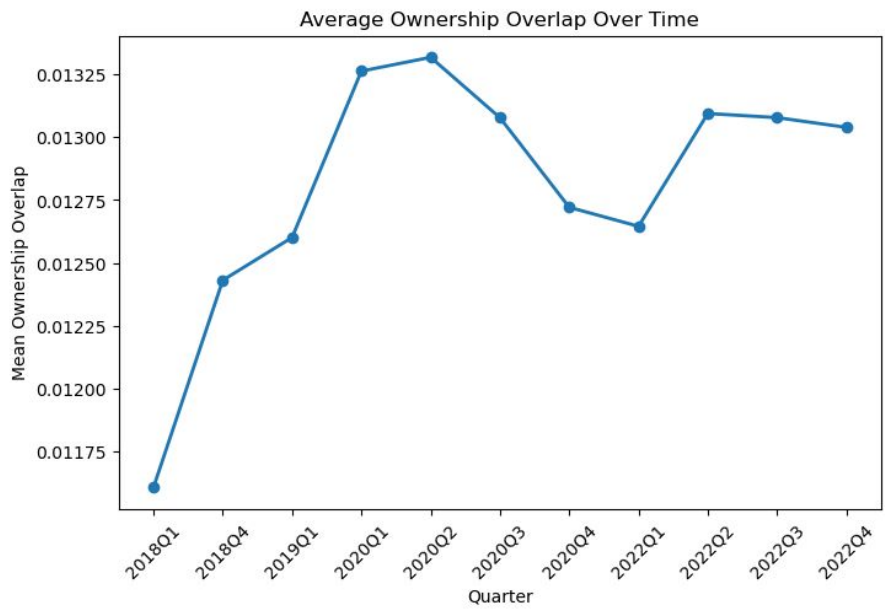
Edges: 171

Density: 1.0

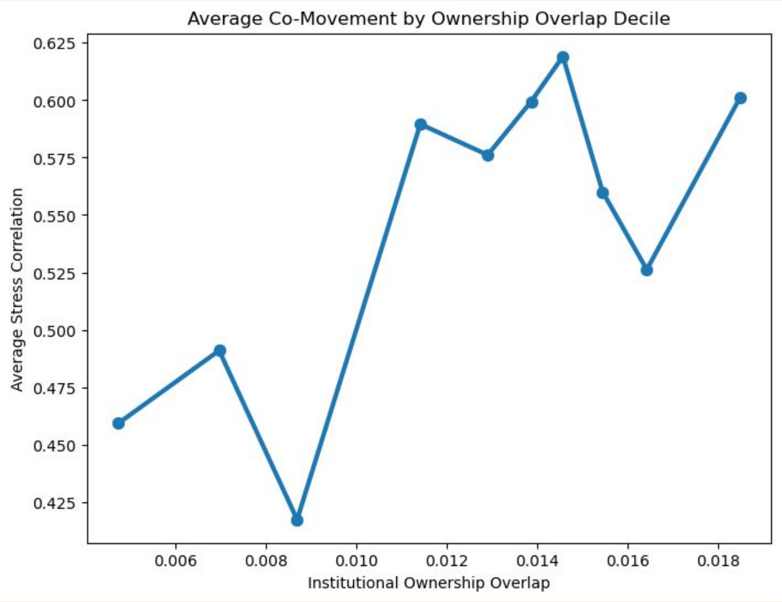
Average Degree: 18.0

Average Weighted Degree: 0.23871735771707234

Average Clustering: 0.40476672433927213



Ownership Overlap and Co-Movement



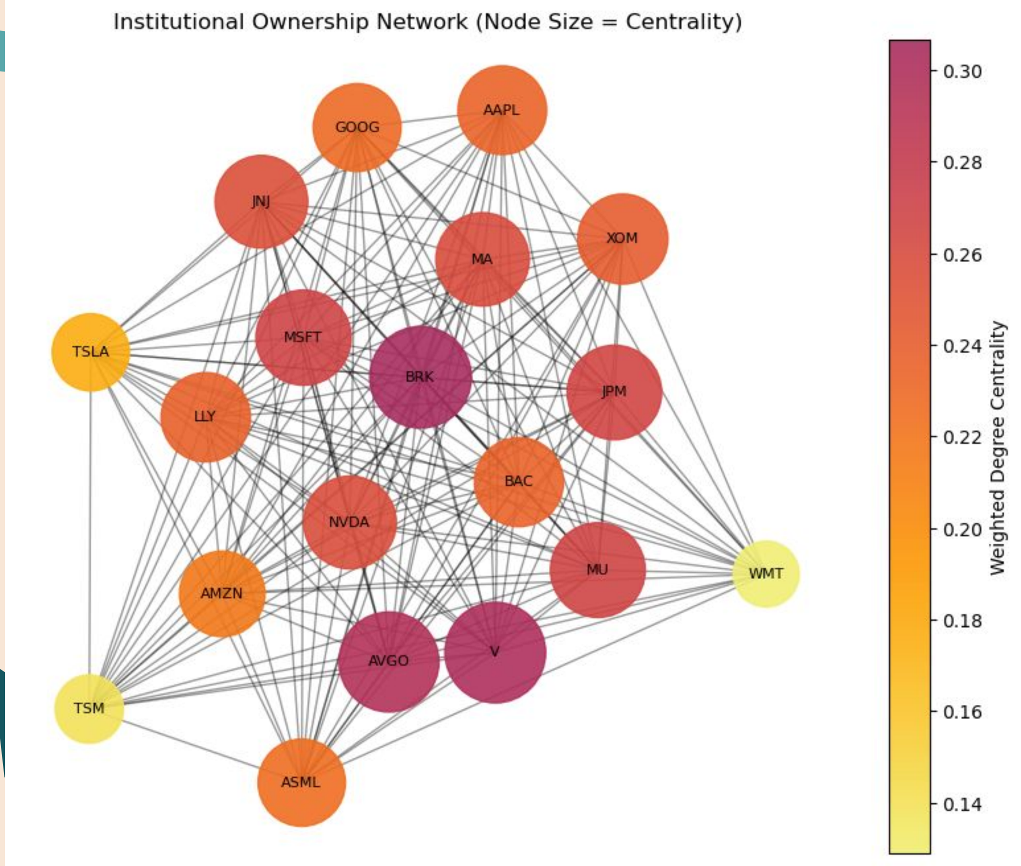
$$\text{StressCorrelation}_{ij} = 0.364 + 11.774^{***}\text{OwnershipOverlap}_{ij} + 0.217^{***}\text{SameSector}_{ij} + \varepsilon_{ij}$$

Variable	Coefficient (Std. Error)
Ownership Overlap	11.774*** (1.206)
Same Sector	0.217*** (0.010)
Constant	0.364*** (0.017)
Observations	3498
R^2	0.093
Adj. R^2	0.092

Robust standard errors (HC3). $^{***}p < 0.01$

- Ownership overlap coefficient: 11.774 ($p < 0.01$)
  - A 1% increase in shared institutional ownership is associated with roughly a 0.117 increase in return correlation
  - Firm pairs with more shared institutional investors tend to have stronger stress correlations
- Same sector coefficient: 0.217 ($p < 0.01$)
  - Firms in the same industry also show higher co-movement, consistent with standard industry effects
- Firm pairs in higher overlap deciles have higher return correlations

Centrality Network – 2020Q1



What it shows

- Network of firms connected through shared institutional ownership for one quarter
- Node size: weighted degree (strength of ownership connections)
- Node color: weighted degree centrality where darker nodes represent firms with stronger ownership overlap with many other firms

How it was built

- Constructed a weighted firm network where edges represent ownership overlap
- Calculated weighted degree as the sum of edge weights for each firm
- Scaled node size as weighted_degree × 10,000
- Colored nodes using an Inferno colormap based on weighted degree
- Positioned nodes using a spring layout
- Visualized with NetworkX and Matplotlib

Descriptive Measures

Centrality Summary Statistics

mean	0.238717
std	0.048259
min	0.128943
25%	0.227322
50%	0.240832
75%	0.266027
max	0.306666

Centrality Distribution Measures

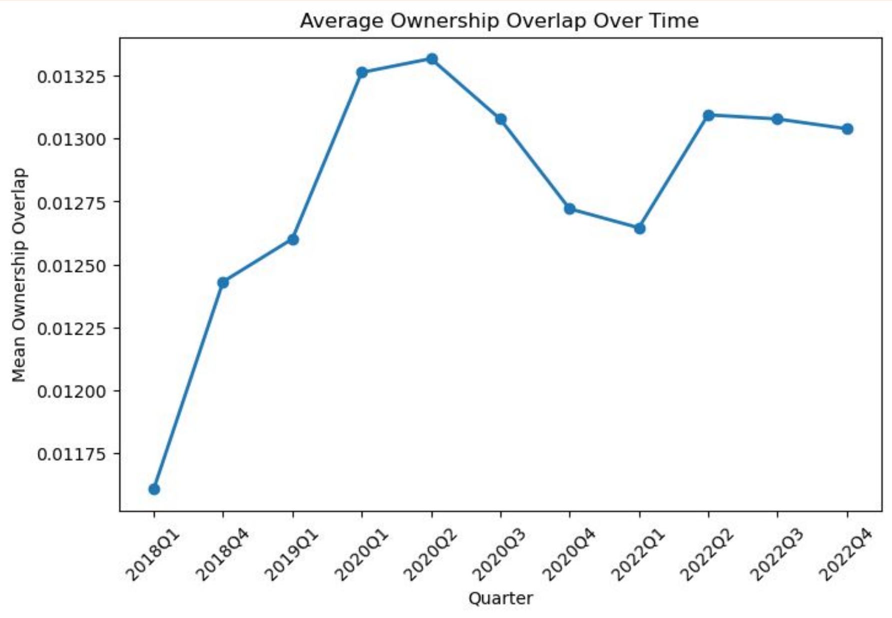
Median Centrality: 0.24083245322732258

Centrality Standard Deviation: 0.04825858866663514

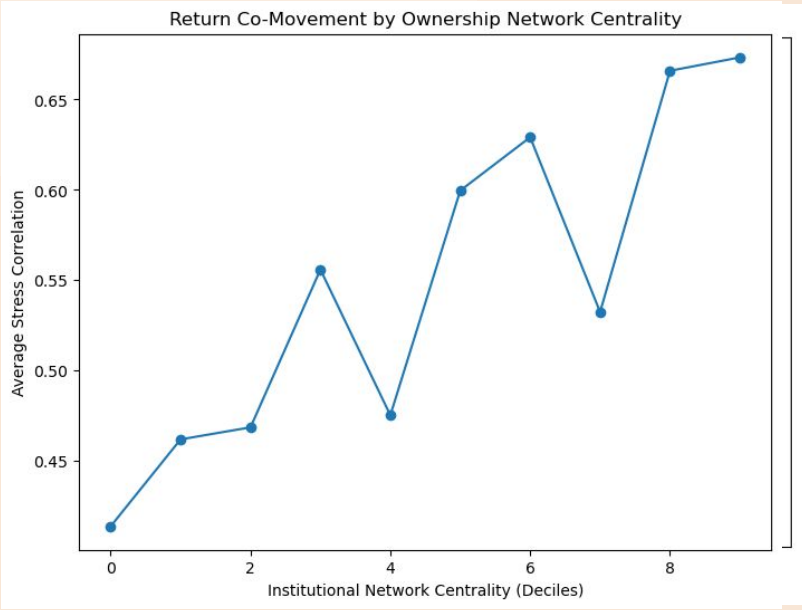
Network Centrality Statistics

Maximum Centrality: 0.30666600906871516

Minimum Centrality: 0.1289434311361574



Network Centrality Regression



$$\text{StressCorrelation}_{ij} = 0.045 + 2.017^{***}\ \text{PairCentrality}_{ij} + 0.208^{***}\ \text{SameSector}_{ij} + \varepsilon_{ij}$$

Variable	Coefficient (Std. Error)
Pair Centrality	2.017^{***} (0.154)
Same Sector	0.208^{***} (0.010)
Constant	0.045 (0.038)
Observations	3384
R^2	0.115
Adj. R^2	0.115

Robust standard errors (HC3). $^{***}p < 0.01$

- Pair centrality coefficient: 2.017 ($p < 0.01$)
  - Since the centrality values range from .05 - .2, we can interpret it to say a 0.10 increase in network centrality is associated with about a 0.24 increase in return correlation during stress periods
  - Firm pairs with higher network centrality tend to have stronger stress correlations
- Same sector coefficient: 0.208 ($p < 0.01$)
  - Firms operating in the same industry exhibit higher return co-movement
- Firm pairs in higher centrality deciles show higher average stress correlations

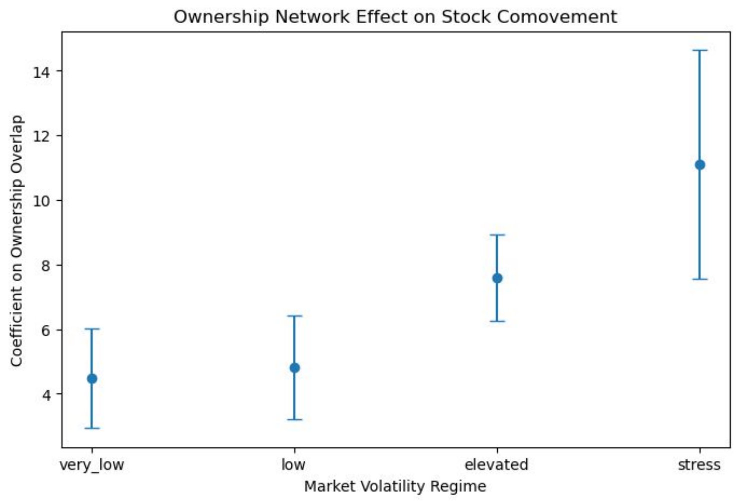
Variations in Volatility Levels

Table 1: Institutional Ownership Overlap and Stock Return Comovement Across Volatility Regimes

	Very Low Volatility	Low Volatility	Elevated Volatility	Stress
Ownership Overlap	4.472***	4.810***	7.585***	11.108***
	(1.532)	(1.601)	(1.341)	(3.535)
Same Sector	0.249***	0.294***	0.208***	0.136***
	(0.019)	(0.020)	(0.017)	(0.042)
Constant	-0.038	0.166***	0.424***	0.472***
	(0.021)	(0.022)	(0.018)	(0.048)
Observations	1939	1939	1939	1578
R^2	0.088	0.108	0.093	0.013

Standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1



- Across all volatility regimes, firms with more shared institutional investors exhibit significantly higher return correlations
    a. Ownership overlap increases stock comovement
- The effect strengthens with market volatility. The ownership coefficient rises from about 4.5 in calm markets to over 11 during stress, indicating stronger comovement when volatility is high.

Conclusion

Answer to Research Question:

Firms with greater shared institutional ownership exhibit stronger return comovement, and this effect becomes larger as market volatility increases

Firms that are more central in the institutional ownership network also show higher correlations with other firms.

Conclusion:
Institutional ownership networks are associated with higher stock comovement, particularly during periods of elevated market volatility.

Limitations:
Shared ownership may also reflect similar firm characteristics or index inclusion, and sector controls might not be capturing all industry relationships. The analysis also shows correlation rather than causal effects.